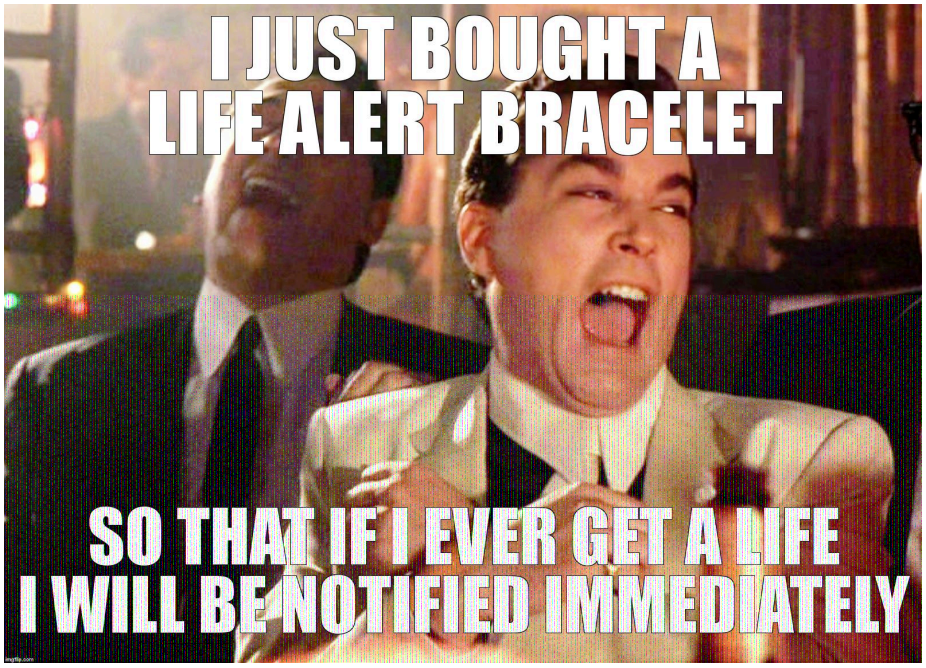
Translating the binary to text for both files, we get “Nothing to see here. Move along.

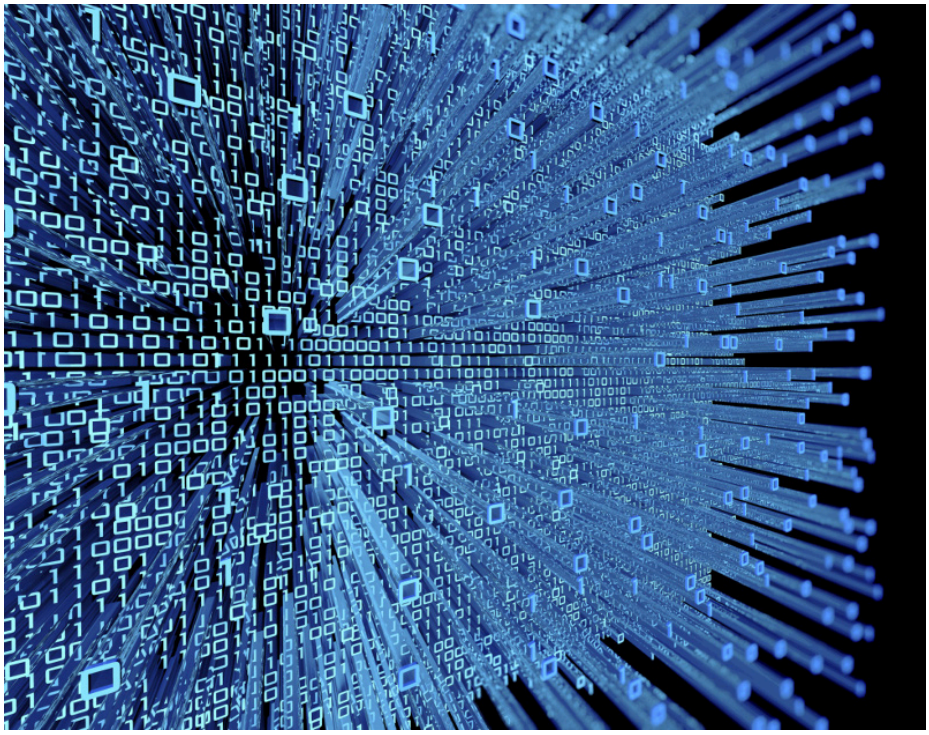
The name of the RVB file translated from base64 gives EPOCH 2023 11 03 00 00 00
The other file that starts with SVA, the name also uses base64 to translate to IP: 138.47.99.31

After editing the timelock program to use the specific epoch given from the file, we were able to log into the ftp server. From this a key and cipher was obtained.

We then used the XOR program on these files, and got a file that gave an image after converting to a .bmp file.
“Life alert” image:



After decrypting this image using “./steg -r -B -o 2048 -i 8 -w image.bmp“ on the steg program, we got the “cyber” image:



Continuing to use the steg program, we got a text file that appeared to be either base64 or hex, but attempting to convert it got us nothing. We were using the program wrong.

Then we used the bit-mode steganography with offset 513 and interval 2 to extract the Constitution text (command - ./steg -r -b -o 513 -i 2 -w result > constitution.txt

After that we discovered another hidden message inside constitution.txt. Used byte mode with offset 32 and interval 64 and use the command (./steg -r -B -o 32 -i 64 -w constitution.txt > final_message.txt and then cat final_message.txt) and recovered the final passphrase which is

PLUTO ROAMS THE KUIPER BELT. Demoted but not forgotten, it wears a heart of ice and nitrogen.

Contributions:

Amanda Berg:

- Edited timelock program so we could get the correct password for the ftp server
- Worked on decrypting image files using steg program
- Documentation

Ryan Bourge:

- Found the IP and EPOCH that translated from Base64 from the file names that were initially given
- Used the Steg program to find the image hidden in the image

Andrew Miletello:

- Found that you needed to add .bmp to file from the xor to get first image to get lifealert image
- Remembered to submit final message in discord
- Worked on decrypting files from ftp server

Saurav Shah

- Extracted the constitution text using bit mode offset 513 and interval 2.
- Discovered the third hidden message inside constitution.txt using byte mode offset 32 and interval 64.
- Successfully recovered the final secret passphrase: PLUTO ROAMS THE KUIPER BELT.

Hayden Mersky

- FTP'd into server and downloaded k3y and c1ph3r files
- Used XOR program to get lifealert file, then converted to bmp
- Used Steganography program to extract cyber image from lifealert
- Used Steganography program to extract Constitution text from lifealert